AP CALCULUS AB

Unit 1 • Lesson 1.15 practice

Lesson 1.15 • Connecting Limits at Infinity and Horizontal Asymptotes

4 multiple-choice questions and 1 free-response question

Question 1

$\lim_{x\to\infty} \dfrac{3x^2+1}{x^2-5}$ is

(A) 3

(B) $-\dfrac{1}{5}$

(C) 0

(D) ∞

Question 2

$\lim_{x\to\infty} \dfrac{x+1}{x^2}$ is

(A) 1

(B) 0

(C) ∞

(D) $\dfrac{1}{2}$

Question 3

$\lim_{x\to\infty} e^{-x}$ is

(A) 1

(B) ∞

(C) 0

(D) $-\infty$

Question 4

$\lim_{x\to\infty} \dfrac{x^2}{e^x}$ is

(A) ∞

(B) 1

(C) e

(D) 0

Question 5 (Free response)

Let $f(x) = \dfrac{4x}{\sqrt{9x^2+1}}$.

(a) Find $\lim_{x\to\infty} f(x)$.

(b) Find $\lim_{x\to-\infty} f(x)$. Show why the sign changes.

(c) Write the equations of all horizontal asymptotes of f.

(d) Find $\lim_{x\to\infty} \dfrac{2e^x+5}{e^x-3}$.